



Robinhood, Reddit, and the news: The impact of traditional and social media on retail investor trading

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ABSTRACT

We study the impact of social media posts and news articles on retail trading on Robinhood by analyzing the net buying activity before and after a (social) media publication. Our findings indicate that both publication types lead to an increase of holders. However, both effects are short-lived, and the effect of social media posts on retail trading is significantly higher. Furthermore, investors buy stocks that are recommended on Reddit regardless of text sentiment. However, they act as contrarians by selling stocks with positive news coverage and vice versa. We thus find evidence for both attention- and information-driven trading behavior.

1. Introduction

In recent years, disruptive technologies, such as social media and commission-free trading platforms, fundamentally changed the way retail investors access financial markets and interact with each other. This has attracted a new generation of retail investors, as evidenced by the doubling of the share of retail traders in U.S. equities over the past decade to more than 20% by 2020 (Tabb, 2021). This increase can be partly attributed to new and easy-to-use trading platforms such as Robinhood, which enable even small and inexperienced investors to participate in financial markets. Another contributing factor is the so-called meme stock phenomenon triggered by discussions on social media platforms like Reddit. Our study contributes to the growing body of literature that deals with the behavior and motives of the emerging group of retail investors that use Robinhood. Specifically, we aim to provide insight into the relative importance of news transmitted through social media versus news conveyed through traditional media in retail investor decision making.

To this end, we utilize three extensive datasets that include the net buying activity of Robinhood users, Reddit posts from the three most influential finance subreddits, *wallstreetbets*, *investing*, and *stocks*, and news articles from various outlets, such as Reuters, CNBC, and Business Insider. After preprocessing and identifying articles and posts that cover a stock, we perform a sentiment analysis. We then present a first visual overview of the changes of the number of holders of a stock around the publication of a post or article.

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Subsequently, we use linear regression analysis to examine the abnormal change in the number of Robinhood users who buy or sell a stock after a post or article publication.

Our results show that Reddit posts have a significant impact on investors that is significantly greater than that of conventional news articles. This can be partly explained by the differences in reaction to positive and negative news. News articles are more likely to have negative sentiment than Reddit posts, which are predominantly positive. However, while investors on average buy the stock mentioned on Reddit regardless of the sentiment, they sell it when it is mentioned in news articles in a positive context and vice versa. This suggests that they behave as contrarians with respect to traditional news, but at the same time pay attention to stocks that are trending on social media. Thus, we can contribute to the discussion between information-based and attention-based trading of retail investors.

The structure of this study is as follows: In Section 2, we review the relevant literature regarding traditional news, social media, and retail trading. In Section 3, we explain the data analysis and processing methods we use and show the descriptive statistics. In Section 4, we present the results. In Section 5, we discuss the implications and limitations of the study. We conclude in Section 6.

2. Background and related literature

In this section, we briefly review the three streams of literature that we use to develop our research idea. In the first two subsections, we summarize research on the impact of traditional news and social media on retail investors and financial markets. Then, we analyze the literature on the behavior of retail investors, specifically Robinhood users. Finally, we draw together these three strands of research and present our research idea.

2.1. Traditional news

Our study is related to several studies that document significant links between news coverage and trading volume or stock prices. Chan (2003) uses a dataset of public headline news to show that stocks that got hit by a price shock continue to go down after bad news, while without news there is a reversal. Tetlock (2007) investigates the *Wall Street Journal*'s "Abreast of the Market" column on U.S. stock market returns and shows that unusually high or low pessimism predicts high market trading volume. Barber and Odean (2008) show that individual investors are net buyers of stocks with features that generate attention, such as stocks in the news. Fang and Peress (2009) study mass media coverage of U.S. companies. They find that stocks with higher media coverage earn lower returns than those with no coverage. This effect is more pronounced among small stocks with high individual ownership. These results indicate that mass media is an important channel for disseminating information to individual investors. Relatedly, Zou et al. (2019) find that stocks without media coverage have significantly higher returns on the Chinese market. Tetlock (2011) analyzes Dow Jones newswire stories and distinguishes between new and stale information (i.e., news that is similar to previous stories). His results indicate that individual investors tend to overreact to stale news, which leads to temporary price movements. Boudoukh et al. (2013) use textual analysis to identify news and calculate sentiment. They find a strong relationship between news and stock price changes. Specifically, market models yield a lower R-squared on news days compared to days without news. This is in line with Roll's (1988) idea that news contains firm-specific information. Therefore, idiosyncratic information is high on news days, which implies that market models that rely on systematic risks have less explanatory power. Furthermore, return variances are higher on news days. Alanyali et al. (2013) find a positive correlation between mentions in the *Financial Times* and transaction volume on the day and the day before. This indicates that news and trading volume are interlinked. Hillert et al. (2014) find that higher coverage in newspapers is associated with stronger momentum, which supports overreaction-based explanations of the momentum effect. Heston and Sinha (2017) use a dataset of more than 900,000 news stories. They find that daily news predicts the stock returns of the next two days, while weekly news has predictive power for one quarter. Furthermore, in contrast to positive news, negative news leads to a delayed reaction.

These studies indicate that traditional news plays an important role in asset pricing. However, in recent years, disruptive technologies have reshaped both the media and brokerage landscape. Specifically, the rise of social media has changed how retail investors interact and discuss strategies, while easy-to-use brokerage platforms facilitate access to stocks for a new generation of retail investors. The major impact of these new technologies was brought to the attention of a wide audience by the short squeeze of the GameStop stock in association with the social media platform Reddit and the trading platform Robinhood. Our examination of these platforms in the period before this short squeeze is therefore of particular interest.

2.2. Social media

There are various social media platforms with different audiences that are used by retail investors to discuss trading strategies and stocks. Several studies use data from Twitter, and sentiment scores in particular. Li et al. (2017) show that sentiment data from Twitter can be used to improve the prediction of price movement of individual stocks. Broadstock and Zhang (2019) find that sentiment on Twitter on specific stocks and the overall market is significantly related to stock price dynamics. McGurk et al. (2020) show that investor sentiment on Twitter has a significantly positive effect on excess returns. They interpret this as evidence of attention-based trading. Duz Tan and Tas, 2021 find that Twitter activity and sentiment predict trading volume on the next day. According to Chawla et al. (2022), news diffusion on Twitter is significantly related to retail investors' intraday trading. Due to attention-based

trading, bid-ask-spreads are reduced on the day of Twitter posts. However, the effect is completely reverted on the next day. This indicates that Twitter does not provide new information but spreads stale news.

Stocktwits is another social media platform that is similar to Twitter. However, its unique feature is that users tag their posts with the ticker and their sentiment for the stock they are discussing. Thus, Stocktwits sentiment is frequently used in research and has been shown to be related to stock returns and intraday liquidity. For example, Agrawal et al. (2018) show that investor sentiment on StockTwits and Twitter influences stock liquidity. Renault (2017) finds that the change in investor sentiment on Stocktwits in the first half-hour predicts the return of the S&P 500 index in the second half-hour. Similarly, Divernois and Filipović (2023) examine the predictive power of sentiment on Stocktwits for abnormal next-day stock returns. We use Stocktwits sentiment, and specifically the dataset from Divernois and Filipović (2023), for additional robustness checks in Appendix A.

Another social media platform that has been analyzed by recent research is Seeking Alpha. It is crowd-sourced and provides news and analysis on financial markets. Chen et al. (2014) find that posts and comments on Seeking Alpha predict stock returns and earning surprises. Farrell et al. (2022) show that there is more retail trading after a publication on Seeking Alpha. Furthermore, using the editorial delay, they demonstrate that the posts are likely to be informative. However, the authors acknowledge that their results may not generalize to other social media platforms.

In our study, we use social media data from the largest finance-related communities on Reddit: *wallstreetbets*, *investing*, and *stocks*. Reddit was founded in 2005 and regularly ranks among the most visited websites worldwide.¹ It is divided into self-created communities, so-called subreddits. What sets Reddit apart from other social media platforms is that members do not follow individuals, but rather these subreddits, which are communities formed based on shared interests. This means that posts on one of the major financial subreddits are read by potentially millions of investors and can therefore have a significant impact on the stock market.

Due to its size and unique design, Reddit, and in particular *wallstreetbets*, have been investigated in recent research. Bradley et al. (2021) consider due diligence reports on *wallstreetbets*. They find that these predict returns before the short squeeze of GameStop but have no predictive power after it. Betzer and Harries (2022) investigate Reddit posts and GameStop trading activity. They find a significant relationship that appears to be attention-based. Lyócsa et al. (2022) investigate the short squeezes of GME, AMC, BB, and NOK and show that posts on *wallstreetbets* are linked to price variations on the next day. Huang and Shum Nolan (2023) use Reddit and Robinhood data. They find that the top three stocks on Reddit outperform the market according to the four-factor CAPM. Hu et al. (2023) find that so-called fanatic and rational agents significantly influence the opinions of naïve investors on Reddit. Furthermore, activity on Reddit predicts stock price movement, trading volume, volatility, retail trading, and deters shorting flows. Chacon et al. (2023) show that there is a significant relation between recommendations on *wallstreetbets* and abnormal trading volume. However, trading strategies based on recommendations are not profitable as they lead to insignificant alphas. Relatedly, Reichenbach and Walther (2023) demonstrate that following stock recommendations on Reddit yields portfolios that would be favored (compared to the market) by investors behaving according to cumulative prospect theory. This indicates that Reddit posts contain information about lottery-like features of stocks and can help to identify interesting betting targets.

In contrast to the majority of studies, we investigate a large number of stocks over an entire period before the short squeeze of GameStop. Furthermore, our dataset of Robinhood holdings enables us to distinguish between net buying and selling, that is to study the direction of trades. We do not only consider immediate effects but consider several days around a post on Reddit. Most importantly, we are among the first to analyze the relative importance of traditional and social media, in particular Reddit, for the trading behavior of retail investors on Robinhood. The study closest to us in this context is Jiao et al. (2020). However, their sample covers the years 2009–2014 and the social media buzz according to the Thomson Reuters MarketPsych Index database, which is the time before new types of social media and trading platforms became popular. Furthermore, they investigate trading volume and variance, while we analyze the changes of stock holdings of retail investors. Our study therefore represents a novel contribution to our understanding of the motives and decision processes of retail investors.

2.3. Robinhood

Our study contributes to the growing stream of literature on retail investors. Many of these recent studies, like us, use data from Robinhood. The results regarding investor sophistication are mixed. On the one hand, Welch (2022) finds that that between mid-2018 and mid-2020 the Robinhood portfolio had good timing and produced positive alphas. For a similar period, Fedyk (2022) shows that the Robinhood portfolio outperforms the market according to various specification of the CAPM. On the other hand, Eaton et al. (2022) investigate outages of Robinhood and traditional brokers. They find that outages of Robinhood are associated with decreased market order imbalances (i.e., increased liquidity), while outages of traditional brokers reduce liquidity. This indicates that investors on Robinhood are rather unsophisticated and likely to take part in attention-based herding. Barber et al. (2022) also provide evidence in favor of attention-induced herding. In particular, they show that after Robinhood outages, the reduction of retail trading is especially pronounced among high-attention stocks. This herding is often followed by negative average returns of up to 20%. Our study contributes to this discussion by analyzing the retail investors' reactions to two different types of media, namely social media and traditional media, that are major channels for generating attention.

Pagano et al. (2021), on the other hand, show that investors on Robinhood engage in both momentum and contrarian strategies. Furthermore, they find that retail investors' impact on market quality, such as price discovery and liquidity, depends on market

¹ For example, in November 2022, it was ranked 8 with 4.82 billion total visits (<https://www.statista.com/statistics/1201880/most-visited-websites-worldwide/>).

conditions. Specifically, they find that investors on Robinhood have a negative effect on market quality during the Covid-19 crisis, while they have a positive effect otherwise. Our study provides insight into the heterogeneous behavior of retail investors by investigating whether news articles or social media posts are a trigger of momentum or contrarian trading strategies.

Other studies also consider the effect of media on retail trading. However, to the best of our knowledge, we are among the first to compare social media to traditional media in this context. [Mohr \(2022\)](#) investigates the impact of outages on Twitter and Facebook but not Reddit on trading on Robinhood. He finds that after outages, trading volume increases, mainly because of selling stocks and closing positions. He concludes that social media distracts investors and does not provide new information. [Fedyk \(2022\)](#) finds that Robinhood trading can best be explained by attention-based trading (e.g., after extreme returns), a “buy-the-dip” effect that favors large and well-known companies that have incurred high stock price losses, and positive sentiment on the subreddit *wallstreetbets*. However, in contrast to us, only one subreddit, and only the next trading day are considered. Furthermore, the effects of Reddit are not compared to traditional media, which is a major contribution of our study.

2.4. Research idea

Disruptive technologies have drastically changed the media and brokerage landscape. Our analysis of the behavior of a new generation of retail investors with respect to both traditional and social media contributes to this timely topic by investigating whether retail investors pay more attention to social media posts or traditional news articles. In addition, unlike many studies, we do not consider the potential increase in trading volume but can directly measure whether users are on average net buyers or net sellers of the stock. Furthermore, we consider abnormal changes of the number of holders on Robinhood (i.e., we control for factors such as weekdays and the recent change in the number of holders before a publication). We also consider whether the effects are only short term or are sustainable. Moreover, we distinguish between positive and negative sentiment and different types of Reddit posts and news articles (i.e., subreddits or publishers). Analyzing the effects of different communities or publishers gives an indication of the level of sophistication of investors, as it shows whether investors favor sources that are likely to include original and business-relevant news. In this way, we contribute to the discussion of whether trading of retail investors is based on information or on attention and herding.

3. Data and methodology

3.1. Data sources

To be able to examine our research question, we need to compile three datasets: a dataset of the trading behavior of retail investors, a dataset of social media posts, and a dataset of news articles.

For retail investor trading behavior, we rely on a unique dataset from Robinhood, a prominent commission-free trading platform. Between 2018 and 2020, Robinhood provided an Application Programming Interface (API). To compile our dataset, we leverage the information made available by [Robintrack.net](#),² a platform that accessed Robinhood’s API approximately once an hour. The dataset includes the stock ticker symbol, the timestamp, and the number of users holding a particular stock at that moment. This dataset sets itself apart because it provides data from a large and relevant group of retail investors, allowing us to examine their behavior. Unlike conventional approaches that rely on retail trading volume, our dataset enables us to calculate the direction of trading (i.e., net buying activity).

For social media posts, we use the Pushshift API ([Baumgartner et al., 2020](#)), which allows us to access historical posts from Reddit. We obtain all posts published within the Robinhood dataset period from the three largest subreddits dedicated to stocks: *r/wallstreetbets*, *r/investing*, and *r/stocks*. We then identify the stocks that can be traded on Robinhood (see next section).

The news articles are obtained from the “All the News 2.0” dataset published by [Thompson et al. \(2020\)](#). Covering the period from January 2016 to March 2020, the dataset contains approximately 27 million scraped news articles. As sources for the articles, we considered 26 different publicists. For each article, the dataset information contains, among other information, the title and text of the article, as well as the date of the publication.

For additional financial data and the full names of companies, we use the Refinitiv Eikon database.

3.2. Data processing

In total, the Robinhood dataset includes 164,350,515 ticker timestamp combinations and 8595 unique stock tickers. The data spans from May 2, 2018 to August 13, 2020. However, the dataset does have some limitations. There are missing data points on specific dates, namely August 9, 2018, January 24 to January 29, 2019, March 3, 2019, and January 7 to January 15, 2020. Additionally, some stocks are absent (see [Welch, 2022](#)).

We perform several steps during the preprocessing phase. Firstly, we remove duplicate entries. Since the data has gaps on certain dates, we employ linear approximation to estimate the number of users holding a stock in such instances. This approach mitigates the impact of API outages and prevents sudden jumps in data that could distort our analysis. Finally, as we examine the daily changes in the number of users holding, we extract only the last data point for each day.

² See <https://robintrack.net/data-download>.

Next, we prepare the Reddit dataset. Firstly, we exclude posts that have been automatically deleted due to violations of community rules. Secondly, as Reddit posts commonly use stock tickers rather than the full stock names, our focus lies in identifying these tickers within the post titles. To achieve this, we extract all unique tickers from the Robinhood dataset, which serve as a reference. However, to avoid false positives, these identified tickers are then cross-referenced with a list of English words.³ We only accept ticker mentions that are present in the word list or are common acronyms, such as YOLO, if they are preceded by a "\$" sign in the title of the post. Finally, to quantify the popularity of each stock mentioned on Reddit, we calculate the number of mentions for each stock on a daily basis.

The news dataset contains 2,688,878 articles covering the period from January 2016 to March 2020. Since this is shorter than the time frame covered by the dataset for Reddit posts, this upper bound determines the period under review in our study. In the first step of the preprocessing, we drop entries with missing values for the article title. Afterwards, we split the titles and articles into sentences. In order to match stocks to news articles, we apply named entity recognition to the articles.⁴ We employ the Flair⁵ framework (Akbik et al., 2019; Schweter and Akbik, 2020) for this purpose. Thus, named entities are extracted from the articles and labeled as person, location, organization or miscellaneous.

We next filter the named entity annotations to only consider organizations. Afterwards, we extract and clean the base name of the annotations. For this purpose, we harmonize or remove organization types, like "Ltd." or "Corp." and unnecessary name extensions, like "holding" or "nasdaq non-national" from the named entity annotations. To account for the renaming of "Facebook" to "Meta" and "Google" to "Alphabet," we replace the annotations accordingly. In addition, we strip the punctuation and convert the annotations to lowercase letters. Finally, we tokenize the annotations and drop the tokens with a length smaller or equal to one.

In addition to the named entity annotations, we collect the company names in different writing styles (e.g., long vs. short) and preprocess them analogously. Finally, in order to assign stock tickers to news articles, we compare company names and named entity annotations by checking for a match token by token. When all tokens from both the company name and the named entity annotation are matched independently of the order, we consider the pair of company name and named entity annotation to be a match. In the case of at least one match between one of the name spellings for a ticker and the named entity annotation, we assign the corresponding ticker to the article.

Regarding the sentiment of Reddit posts, we follow the approach in Reichenbach and Walther (2023, 2024) and use VADER sentiment analysis to calculate sentiment scores based on word lists. We customize the predefined word list to account for the finance-specific meaning of words according to Loughran and McDonald (2011), and the most frequently used words and emojis on Reddit.

We estimate sentiment for the news articles using so-called transformer neuronal networks. In particular, we apply the DistilBERT model architecture developed by Sanh et al. (2019) to the title and text of the news articles. Since the model only supports sequences with a maximum length of 512 tokens, longer articles are truncated accordingly.⁶ To account for the specific vocabulary and tone of financial news, we use a model⁷ that has been fine-tuned using a dataset of financial news sentiments.

3.3. Descriptive statistics

Fig. 1 presents an overview of the trends over time in our three datasets, namely the number of Reddit posts and news articles per day that include a stock and the number of holdings on Robinhood. Since data on news articles are only available until April 1, 2020, we limit our analysis to this period.

The average number of news article that reference a stock from Robinhood fluctuates between 50 and 100 per day and slowly grows over time. On Reddit, the activity was higher in 2018 (~100 per day) than in 2019 (~50) and the number of posts exhibit a strong growth during the Covid-19 pandemic in 2020. The total number of Robinhood holdings increases from around 5 million to 25 million with a notable acceleration of growth in 2020 as well.

Table 1 provides a further breakdown of the sources of news articles and posts in our dataset.

In total, our dataset contains 56,011 news articles and 52,393 Reddit posts that include a stock tradable on Robinhood. The news comes from 26 different news outlets, most of which focus on business or political news. The largest three, Reuters, CNBC, and Business Insider, account for over half of the total number of news stories. For Reddit posts, we include the top three financial subreddit: *wallstreetbets*, *stocks*, and *investing*. Posts from *wallstreetbets* amount to about 80% of all posts. The number of subreddit subscribers grew from 251,519 to 1,406,395 for *wallstreetbets*, from 517,428 to 1,128,054 for *investing*, and from 234,484 to 791,080 for *stocks* during our sample period.

The five companies that received the highest total coverage in the respective medium are shown in Table 2. In addition, the table shows the day and the related event on which the most articles were published for those stocks.

³ Specifically, the GradyAugmented list from the qdapDictionaries R package.

⁴ Since mentions of a company in the article title are more likely to be noticed by investors and are a stronger indicator that the article contains news about the company mentioned, we focus on them in the main analysis. As an additional robustness check, we perform named entity recognition on the first five sentences of the article and report the results in Appendix B. Due to the high computational demand, named entity recognition cannot be applied to the full text of all news articles. Thus, we perform named entity recognition on the full text for a subsample and compare the overlap.

⁵ See <https://huggingface.co/flair/ner-english-fast>.

⁶ As an additional robustness check, we also apply the sentiment model to the article title only and present the results in Appendix C.

⁷ See <https://huggingface.co/mrm8488/distilroberta-finetuned-financial-news-sentiment-analysis>.

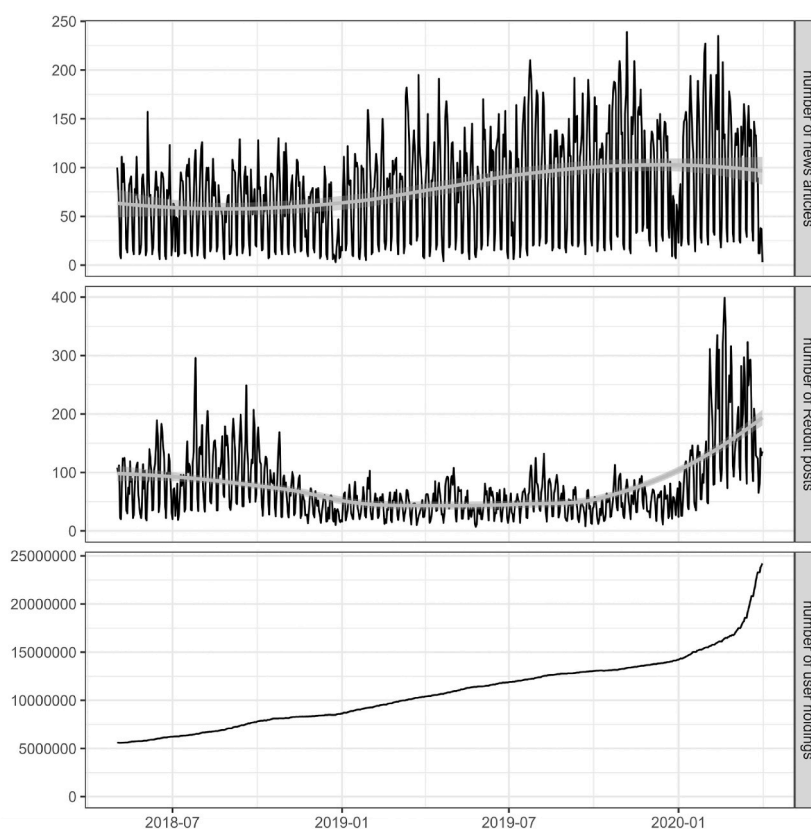


Fig. 1. Overview of the news, Reddit, and Robinhood datasets. The top panel shows the number of news articles per day that include a stock available on Robinhood, and the middle panel shows the number of Reddit posts per day. The third panel depicts the total number of holdings across all stocks on Robinhood. The period shown is from May 2, 2018 to April 1, 2020.

Table 1

Number of publications per news source or subreddit.

News Sources or Subreddits						
News	Reuters	14,879	Vox	824	Vice	227
	CNBC	14,465	Washington Post	799	Hyperallergic	140
	Business Insider	4276	The New York Times	673	Economist	135
	The Verge	3890	Refinery 29	502	Buzzfeed News	102
	Mashable	3810	People	496	Vice News	62
	TechCrunch	3351	Gizmodo	463	TMZ	47
	CNN	2328	Wired	351	New Republic	21
	The Hill	1985	Fox News	290	New Yorker	4
	Axios	1624	Politico	267	Total	56,011
Reddit	wallstreetbets	41,846	stocks	6055	investing	4492
					Total	52,393

Apple is the company most covered in news articles (7168 articles), followed by Amazon (6253), Tesla (4000), Boeing (3364), and Netflix (3065). It is noticeable that on the days with the most mentions, the news is often negative for the company, which is also reflected in a falling share price. For example, the day before November 22, 2019, Tesla presented a new SUV, the Cybertruck. The media focused on the fact that the supposedly unbreakable car windows were shattered during the presentation. Another example is the Boeing 737 MAX crash in Ethiopia on March 10, 2019. There was very strong media interest in the following days, culminating in our dataset on March 12. Due to safety concerns, numerous Boeing aircrafts were grounded, which led to a decline in the share price.

Reddit posts, on the other hand, are more often positive. The most popular stocks are Tesla (4109), AMD (3082), Microsoft (1327), Virgin Galactic (1193), and Amazon (1094). In terms of content, the posts are primarily concerned with changes in stock prices and short squeezes. Although there is usually a trigger, such as AMD having just released new server graphics cards, the discussion on Reddit is mostly focused on the recent stock price development. Note that the period under consideration is before the GameStop short

Table 2

Top 5 stocks in the news and on Reddit. The upper left part of the table shows the stocks that have appeared in the news the most overall. To the right of each of these stocks is listed the day and number of articles on which the company received the most coverage. Also listed is the likely reason for the reporting and the stock's return on that day (measured by using closing prices). The lower part of the table presents the same for Reddit posts.

	Ticker	Description	Total	Date (Maximum)		Event	Return
News	AAPL	Apple	7168	2020-02-18	(103)	Investor update (covid)	−1.8%
	AMZN	Amazon	6253	2019-09-25	(66)	Hardware launch event	+1.5%
	TSLA	Tesla	4000	2019-11-22	(38)	Launch of Cybertruck	−6.2%
	BA	Boeing	3364	2019-03-12	(81)	Plane crash in Ethiopia	−6.1%
	NFLX	Netflix	3065	2019-07-18	(39)	Loss of US subscribers	−10.3%
Reddit	TSLA	Tesla	4109	2020-02-04	(197)	Short squeeze	+13.7%
	AMD	Advanced Micro Devices	3082	2018-08-27	(84)	New hardware, stock price surge	+5.3%
	MSFT	Microsoft	1327	2020-02-11	(88)	FTC investigation	−2.3%
	SPCE	Virgin Galactic	1193	2020-02-18	(153)	Relocation of spaceship, stock price surge	+5.6%
	AMZN	Amazon	1094	2018-07-26	(32)	Earnings after close	−2.9%

Table 3

Number of news articles and Reddit posts with positive, neutral, and negative sentiment.

Sentiment	News		Reddit	
Positive	22,509	38.8%	25,754	49.2%
Neutral	22,720	39.1%	19,191	36.6%
Negative	12,833	22.1%	7448	14.2%
Total	56,011	100%	52,393	100%

squeeze in January 2021 and the emergence of other meme stocks. Interestingly, GameStop is only mentioned 261 times in Reddit posts from May 2018 to April 2020, which shows that the taste of stocks can change fast.

These individual observations regarding the distribution of negative and positive information are also supported by the results of the sentiment analysis presented in Table 3. We find a higher proportion of news articles with negative sentiment (22.1%) compared to negative Reddit posts (14.2%). Conversely, positive sentiment is more prevalent in Reddit posts, with a share of 49.2%, compared to news articles at 38.8%. While the tone of news articles seems to be more balanced between positive and negative sentiment, Reddit posts in these financial subreddits are overwhelmingly positive.

4. Results

4.1. Visual overview

First, we provide a visual overview of the impact that news and Reddit posts have on the number of Robinhood users who own a particular stock. Similar to Welch (2022), we define the number of users holding a stock i on day t as N_t^i . Thus, the absolute change of the number of users is $\Delta N_t^i = N_t^i - N_{t-1}^i$.

Fig. 2 shows the arithmetic mean of the daily change of the number of Robinhood users holding a stock ΔN_t^i under the condition that at least one Reddit post (solid line) or news article (dashed line) was published. The figure includes the 20-day period before and after the event. In addition, the average growth in the month before is shown in gray as a basis for comparison. It is evident that the stocks in both categories are gaining buyers on average. However, the stocks mentioned on Reddit are already gaining more popularity before the post. For both news and Reddit, there is a significant increase in users until the day of the event, when it reaches its maximum and then stabilizes at a higher level than the month before. Stocks discussed on Reddit grow more overall, both before and after the day of the post.

The downside of the plot in Fig. 2 is that it does not consider how many publications have been published. It can be expected that more articles or posts will lead to more attention and consequently more buyers. Therefore, Fig. 3 presents graphs for both variants, with at least one (solid line; identical to Fig. 2), at least two (dashed line), and at least three publications (dotted line) on the day under consideration, respectively. For both news articles and social media posts, the baseline level increases with the number of publications. Furthermore, the increase on the day of the post or article is also higher, and especially pronounced for Reddit posts.

Figs. 2 and 3 provide an overview of the change in holders of a stock in the period before and after the event, given that there was at least one post or article. However, this visual account has some weaknesses because it does not control for other factors that might explain the change in user numbers for a stock. In addition, the number of posts or news cannot be adequately accounted for. As Fig. 3 shows, it is to be expected that more of these lead to more attention and thus to a stronger change in number of users holding the stock. Therefore, we incorporate these findings in our regression models.

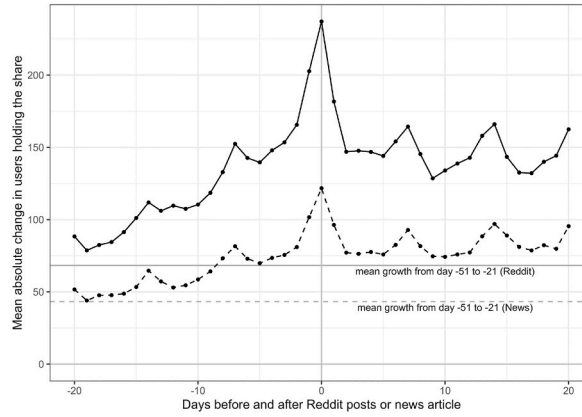


Fig. 2. Mean absolute change in users holding before and after the publication of at least one Reddit post or news article. The figure presents the average daily change in Robinhood users holding a stock mentioned in at least one Reddit posts (solid line) and news articles (dashed line), during a 20-day period surrounding the event. The gray lines indicate the average growth in the month prior.

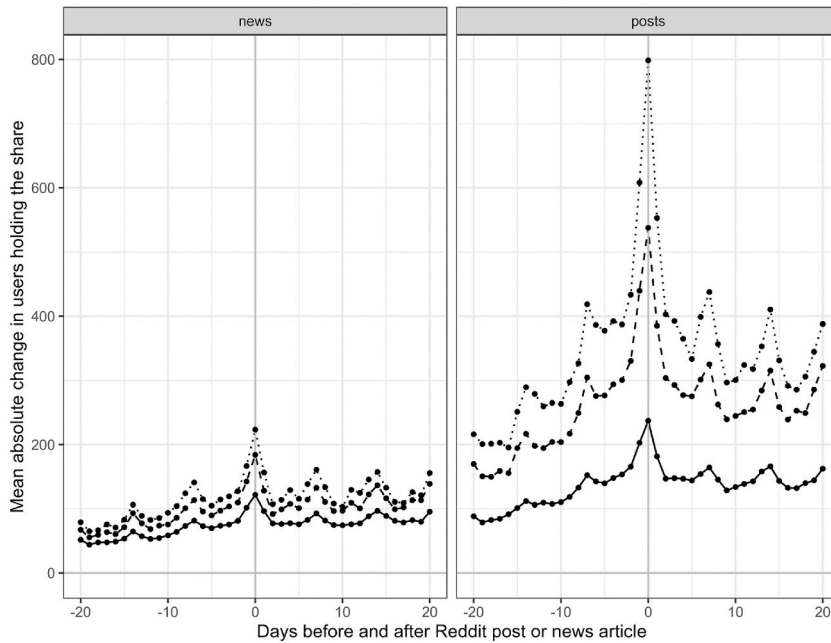


Fig. 3. Mean absolute change in users holding before and after the publication of at least one, two or three Reddit posts or news articles. One or more publications on day 0 is indicated by a solid, two or more by a dashed and three or more by a dotted line.

4.2. Regression models

In order to better understand the impact on retail investors decision-making and address the disadvantages of the approach discussed above, we use a number of regression models:

$$\Delta N_{t+x} = \alpha + \beta_1 \bullet Posts_t + \beta_2 \bullet News_t + \beta_3 \bullet Quarterly\ Report_t + \beta_4 \bullet Decile\ AV_t + \beta_5 \bullet Decile\ Return_t + \beta_j \bullet Controls_{j,t+x} + \varepsilon.$$

In the equation above, the dependent variable ΔN_{t+x} denotes the change in the number of users holding a stock from the end of day $t+x-1$ to day $t+x$, i.e., the net buying activity on day $t+x$. We start with the day of the post or article ($x = 0$) for the first model and track the change of users holding until one trading week after the event ($x = 5$). The intercept is denoted by α , β_1 to β_5 represent regression coefficients, and ε is the error term.

The relevant independent variables are the number of Reddit posts on day t ($Posts_t$) and the number of news articles on day t ($News_t$). Furthermore, we control for the potential effect of an earnings announcement on day t (dummy variable $Quarterly\ Report_t$). Following Barber and Odean (2008), we also account for abnormal trading volume (AV) and extreme one-day returns from day $t-1$ to

Table 4

Linear regression models 1a (base model): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t + x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to Newey and West (1987, 1994). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	−3.13 ***	−2.90 ***	−2.79 ***	−2.85 ***	−3.00 ***	−3.08 ***
Posts	57.98 ***	12.60	−5.66	−3.00	−0.79	−2.35
News	6.54.	2.00	0.40	0.71	1.15	0.01
Quarterly Report	16.15 ***	2.98.	−3.18 ***	−2.67 ***	−2.75 ***	−4.28 ***
AV (1st Decile)	−1.30	0.57	1.11 *	0.92 *	0.73	1.02
AV (10th Decile)	23.82 ***	6.47 ***	−0.60	−0.06	−1.05	−0.75
Return (1st Decile)	3.85 ***	0.36	0.61	−0.69	0.65	−0.49
Return (10th Decile)	4.03 **	3.32 **	−0.56	0.89	−1.60 *	−2.16 **
Mean Day	0.73 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.18 ***	1.23 ***	1.26 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.231	0.209	0.208	0.208	0.208	0.208

day t that may result in increased attention and thus retail trading activity. We measure *AV* as the volume on day t divided by the mean trading volume over the previous 252 trading days. We then include 10 dummy variables for each decile of both variables and an additional one if there are no return or volume data available.

Furthermore, as we are primarily interested in the *abnormal* change of the number of retail investors, we need to control for several additional factors that might influence the net buying activity on Robinhood. These are the year and day of the week of day $t + x$, the arithmetic mean of the change of users holding across all stocks on day $t + x$ (*Mean Day*), and the arithmetic mean of the change of users holding the considered stock in the month before the event (i.e., from day $t + x - 21$ to day $t + x - 1$: *Mean Month*). The rationale behind this approach is to measure the unexpected change in the number of users holding after a Reddit post or news article. For example, if a stock has been gaining an average of 500 new holders per day over the past month, it is more likely that it will show positive growth on the day of a news article, even if there is no causal relationship between the two. The same is true if the average number of users across all stocks increases on the day of the event because Robinhood's user base is growing due to external factors.

Table 4 presents the first models with the net change in the number of holdings ΔN_{t+x} as the dependent variable. On the day of a Reddit post, the abnormal change in the number of holdings is +57.98 per post and +12.60 per post on the next day. On the following days, the number of holdings is reducing (insignificantly). For traditional media articles, the number of additional holdings is around nine times smaller on the day of the article (+6.54 users holding) and the effect quickly reduces in the following days but does not become negative. Earnings announcements also result in a positive abnormal net buying activity on the first and second day, but after day two, the interest of Robinhood investors seems to decrease, which reduces the number of holders. In line with Barber and Odean (2008), high abnormal volume and high or low one-day stock returns lead to a short-term increase of buyers.⁸ Note that we calculate robust standard errors by controlling for heteroscedasticity and autocorrelation according to Newey and West (1987, 1994).

Alternatively, as in the previous subsection and as a robustness check, we perform the same analysis with the *Posts* and *News* variables as dummy variables (1 if a Reddit post or news article was published on that day; 0 otherwise). However, the results in Table 5 are qualitatively similar to the results in Table 4.

Looking at sentiment (Table 6), the difference in the strength of the effects of news articles and Reddit posts can be partially explained by the fact that the reaction to positive and negative narratives is different.

Specifically, the analysis reveals that for Reddit posts, a stronger positive sentiment is associated with a greater influx of new stockholders on Robinhood on the initial day. By contrast, this pattern is reversed for news articles, where a more positive sentiment leads to a reduction in users holding. Interestingly, investors tend to increase their holdings when confronted with negative news on average but opt to sell when confronted with positive news.

Next, in Table 7, we break down the sources of the news articles and the social media posts in more detail.

A post in the *stocks* subreddit has by far the largest impact, followed by posts from *wallstreetbets*. For the latter, however, the growth drops off quickly, while for *stocks*, a significant number of new buyers is still added the next day. In terms of news, the positive increase in holdings in Table 4 is mainly driven by Reuters reports.

5. Discussion

Our results suggest that retail investors' trading decisions can partly be described as information-driven and attention-driven behavior. Both Reddit posts and news articles have a positive impact on net buying activity on Robinhood on the publication day. Additionally, the growth is higher when more articles or posts are published. However, the effect of a Reddit post is significantly larger

⁸ Note that we control for all 10 deciles of *AV* and the one-day return, but report only the top and bottom deciles of both variables, as they are the most important and have been shown in other studies to impact trading volume (e.g., Barber and Odean, 2008). The other deciles are almost always insignificant in all models.

Table 5

Linear regression models 1b (dummy variables): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t + x$, with x ranging from 0 to 5. Independent variables of interest are binary variables indicating whether a Reddit post or news article was published on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	−3.25 ***	−3.02 ***	−2.86 ***	−2.92 ***	−3.04 ***	−3.10 ***
Posts (dummy)	105.25 ***	44.72 ***	5.71	7.96	5.00	0.03
News (dummy)	26.59 **	12.53	0.10	−4.31	−1.54	−2.41
Quarterly Report	17.85 ***	2.51	−3.77 ***	−2.91 ***	−2.74 ***	−4.40 ***
AV (1st Decile)	−1.28.	0.76	1.22 *	0.98 *	0.75	1.03.
AV (10th Decile)	25.22 ***	6.50 ***	−0.88	−0.22	−1.09	−0.84
Return (1st Decile)	4.16 ***	0.25	0.48	−0.77	0.62	−0.52
Return (10th Decile)	4.71 ***	3.23 ***	−0.77	0.77	−1.64 *	−2.21 **
Mean Day	0.71 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.21 ***	1.23 ***	1.25 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.214	0.209	0.208	0.208	0.208	0.208

Table 6

Linear regression models 2 (sentiment): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t + x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles, divided by their sentiment, on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	−3.10 ***	−2.91 ***	−2.80 ***	−2.85 ***	−3.00 ***	−3.07 ***
Posts (positive)	75.32 ***	5.66	−6.89	−3.06	−2.88	5.06
Posts (neutral)	48.16 *	16.99	−21.20 *	−16.53 *	−24.72 ***	−21.79 *
Posts (negative)	27.16	26.09	46.12.	37.94	76.69 *	27.62
News (positive)	−17.20.	−10.62.	−2.95	−2.07	2.74	2.08
News (neutral)	−0.07	1.55	2.85	0.94	−2.04	−4.27
News (negative)	50.34 ***	19.60.	−0.37	3.08	2.21	3.38
Quarterly Report	15.20 ***	3.31.	−2.71 **	−2.48 ***	−2.72 **	−4.79 ***
AV (1st Decile)	−1.34 *	0.56	1.16 **	0.96 *	0.81	1.06.
AV (10th Decile)	23.60 ***	6.42 ***	−0.52	−0.02	−1.02	−0.78
Return (1st Decile)	3.60 ***	0.17	0.48	−0.81	0.45	−0.58
Return (10th Decile)	4.35 ***	3.47 ***	−0.54	0.91 *	−1.62 *	−2.18 **
Mean Day	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.18 ***	1.23 ***	1.26 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.233	0.210	0.209	0.208	0.209	0.208

Table 7

Linear regression models 3 (subreddits and news sources): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t + x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles, divided by their sentiment, on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	−3.13 ***	−2.92 ***	−2.79 ***	−2.86 ***	−3.00 ***	−3.10 ***
Posts (Wallstreetbets)	52.17 ***	8.49	−5.55	−3.86	−0.69	−4.26 ***
Posts (Stocks)	157.29 ***	56.42 *	−7.54	7.58	−9.73	2.75
Posts (Investing)	37.95	48.59.	−1.93	1.91	12.46	34.00 *
News (Reuters)	19.24.	23.97.	5.02	7.95	11.26.	10.33.
News (CNBC)	5.63	−16.96 *	−5.09	−2.85	−4.26	−5.10
News (Other)	0.98	1.32	1.04	−0.89	−0.86	−2.31
Quarterly Report	15.43 ***	2.60	−3.20 **	−2.94 ***	−3.12 ***	−4.64 ***
AV (1st Decile)	−1.27.	0.58	1.11 *	0.92 *	0.73	1.02.
AV (10th Decile)	23.53 ***	6.36 ***	−0.59	−0.11	−1.08	−0.83
Return (1st Decile)	3.76 **	0.38	0.62	−0.69	0.67	−0.47
Return (10th Decile)	4.02 **	3.38 ***	−0.55	0.90	−1.59 *	−2.13 **
Mean Day	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.17 ***	1.23 ***	1.26 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.232	0.210	0.208	0.208	0.208	0.208

compared to the effect of a news article, and both quickly become insignificant over the following days. Quarterly earnings reports initially also lead to growth, but this trend weakens and reverses in the following days, indicating that investors react quickly to new information. Furthermore, Robinhood investors buy stocks with high abnormal trading volume and high or low one-day returns, which is in line with previous research on retail trading behavior (Barber and Odean, 2008) and indicates attention-driven behavior.

Analyzing the sentiment of the articles and posts reveals partial explanations for the varying strengths of these effects, as reactions to positive and negative stories differ depending on the type of media. Positive sentiment in Reddit posts attracts the highest growth in users holding, but negative sentiment still leads to investors buying, indicating the influence of social media attention on purchase decisions. Conversely, news articles show an opposite pattern, with positive sentiment leading to a lower number of holders and investors buying after negative news articles, reflecting a contrarian pattern among Robinhood investors ["buy the dip", see for example Fedyk (2022)]. These findings may provide an explanation for why previous studies have shown that retail investors use both momentum and contrarian trading strategies (Eaton et al., 2022; Pagano et al., 2021; Barber et al., 2022; Welch, 2022).

Furthermore, when analyzing posts from different subreddits, it becomes evident that posts in the *stocks* subreddit lead to more net buyers compared to those in *wallstreetbets*. Firstly, it should be noted that there are significantly more posts in *wallstreetbets* and the number of posts a potential investor can read is limited. Thus, it can be assumed that the attention of users is distributed more widely among posts, which reduces the impact per post. In addition, the posts of the subreddits differ significantly in content and tone. While *wallstreetbets* discusses high-risk investments, *stocks* and *investing* focus more on conventional stocks and long-term strategies (Agrawal et al., 2022). Furthermore, the effect of posts from *stocks* and *investing* on new buyer numbers persists longer, supporting the notion of a longer-term orientation among users of these subreddits (Reichenbach and Walther, 2023).

Our study has several limitations. First, the Robinhood dataset only contains information about the number of holders and not the total amount of holdings. However, both measures should be highly correlated. Furthermore, the change in the number of holders is a good proxy of retail investors' interest in a specific stock, as it measures how many investors bought (or sold) a stock after the publication of a post or article.

Second, the social media landscape evolves and grows rapidly. For example, since the end of our sample period in April 2020, the number of redditors on *wallstreetbets* increased by more than 10 times. Therefore, it is likely that the relative importance of social media for retail investors increased. It is conceivable that the new members of the communities have different characteristics from the early adopters of social media that we investigate. Furthermore, we focus on Reddit in our main analysis, which might differ from other platforms in its effect on investors. Thus, as an additional robustness check, we conducted the same analysis as in Section 4 and replaced Reddit with another popular social media platform, Stocktwits (see Appendix A). Using more than 28 million posts, the main results (i.e., greater influence of social media posts compared to news articles and Robinhood investors act as contrarians with respect to news and as trend followers with respect to social media posts) are all supported, which attests to their robustness. However, analyzing the influence of other social media platforms with a potentially different user base such as Twitter or Seeking Alpha remains an interesting avenue for future research.

Third, we only consider the sentiment and not the informational content of publications. Specifically, our analysis does not allow conclusions about whether posts on Reddit or news articles contain new and original information or whether they only disseminate stale news. However, measuring the informational content of a publication is difficult and very demanding in terms of computational power. Furthermore, we do not claim to show why investors react to posts or articles or whether their behavior is reasonable but provide insight on which type of media retail investors use more. We show that posts on social media have a significantly stronger effect than traditional articles. However, it is an interesting question for future research to examine the extent to which social media's ability to manipulate retail investor behavior poses a threat and needs to be regulated.

6. Conclusion

Our study aims to improve our understanding of retail investor behavior and decision-making processes in the era of social media and to contribute to the debate on attention-based and information-based retail trading. We use an extensive dataset of Reddit posts, news articles, and trading activity from Robinhood users, and provide evidence that both traditional and social media influence retail investors' trading decisions. Furthermore, our results indicate that the role of social media in the decision-making of retail investors is significantly larger than the role of traditional news articles.

However, the effect strengths differ between subreddits and different sources of news articles. Posts on subreddits that focus on long-term strategies have larger effects on the number of holders indicating a certain degree of sophistication of the considered retail investors.

Furthermore, the sentiment analysis of posts and articles provides insight on the seemingly contractionary observation of recent research that retail investors engage in both momentum and contrarian strategies. Posts on Reddit (and Stocktwits) are overwhelmingly positive and positive posts lead significantly more buyers than sellers on the day of the publication. This indicates that social media posts contribute to momentum strategies and attention-based trading. By contrast, negative news articles have a positive effect on the number of holders on Robinhood while positive news leads to selling behavior. This implies that traditional news contributes to the contrarian behavior of retail investors.

A promising area for future research remains the closer examination not only of the sentiment but also of the content of social media posts and traditional news articles. In particular, it is of great interest how news and narratives spread and how they influence the decisions of private investors and thus ultimately stock prices.

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CRediT authorship contribution statement

Markus Münster: Writing – review & editing, Software, Methodology, Data curation, Conceptualization. **Felix Reichenbach:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Martin Walther:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no competing interests.

Data availability

The authors do not have permission to share data.

Appendix. Additional robustness checks

A. Stocktwits instead of Reddit

As an additional robustness check, we substitute Reddit posts by Stocktwits posts. The main difference between the two social media platforms is that while Reddit is organized in communities (subreddits), which users can become members of, Stocktwits posts are organized by the stock ticker of the discussed stocks and users can follow tickers or other users. This means that posts on Reddit are rarer and the number of users reading a post is likely far greater (e.g., *wallstreetbets* alone had 1.4 million members at the end of our sample period). Furthermore, Stocktwits allows users to label their posts as either “bullish” or “bearish”. This removes the need to perform sentiment analysis and named entity recognition. We use the dataset of [Divernois and Filipović \(2023\)](#) to calculate the number of Stocktwits posts per day and stock ticker and match these with our Robinhood dataset. In total, the dataset contains 28.4 million posts for the period considered, of which 31% are labeled by users as “bullish” and 8% as “bearish,” a similar ratio as in the Reddit dataset.

[Table A1](#), which is the equivalent to [Table 4](#), presents the results for the base models. As the Stocktwits dataset only includes data up to March 23, 2020, the sample size is slightly smaller than for our Reddit models in [Section 4](#).

As in [Table 4](#), social media posts have a significant positive effect on the number of buyers (17.99 additional users per 100 Stocktwits posts on day 0 in [Table A1](#)), but this effect disappears within two days. News articles, on the other hand, have no significant effect. This supports our finding that trading by Robinhood investors is primarily driven by social media and not by news.

Regarding the sentiment of posts and articles, the results in [Table A2](#) are again similar to those in [Section 4](#) (see [Table 6](#)).

We find that bullish Stocktwits posts are associated with a significant short-term increase in users holding, while bearish and neutral posts have no significant short-term effect. Furthermore, as in [Table 6](#), users buy stocks on negative news and sell stocks on positive news. These results support our finding that Robinhood investors act as trend followers with respect to social media posts and as contrarians with respect to news.

B. Rationale of applying named entity recognition on article titles

In the main analysis, we use the titles of news articles to identify stocks that are in the news. This focus on the title is appropriate for two reasons: titles are more likely to be read or seen by more Robinhood users than the entire text, and a mention in the title indicates that the company is an important element of the news article, meaning that the information provided is likely to be relevant to the covered stock. Thus, there is a trade-off between relevance for the stock and the quantity of identified stocks in news articles. Flagging companies that are mentioned later in the text is likely to reduce the average effect of a news article on buying volume on Robinhood. This is confirmed when performing named entity recognition on the first five sentences of the article text in addition to the article title and rerunning the main analysis. [Table B1](#) depicts the corresponding regression results.

[Table B1](#) shows that the number of mentions in the first sentences of the news articles has no significant effect on net buying activity. As our main finding is that investors on Robinhood respond more strongly to posts on social media, considering mentions in the news headline and first five sentences provides additional support for this result.

Furthermore, we also perform named entity recognition on the full text of a random sample of 50,000 articles. The overlap with the previously identified stock tickers amounts to approximately 40%. This confirms that the beginning of the article already covers a large proportion of the stocks mentioned in the whole article. In addition, approximately 28% of the missing matches belong to “Twitter” and “Facebook” which are often cited as the source of a quote or image. This further supports the idea that if the article contains relevant information about the stock, the name of stock is mentioned early.

C. Sentiment from news article titles only

In the main analysis, we perform named entity recognition on the title of the news articles and sentiment analysis on the full text. To better incorporate finance-specific vocabulary the model of [Sanh et al. \(2019\)](#) was fine-tuned on single sentences rather than long, complex texts. For these reasons, in an additional robustness check, we use the headlines to estimate the sentiment for the whole news article. [Table C1](#) shows the regression results for the adjusted methodology. They are similar to the results in [Table 6](#) and confirm that the findings of the main analysis are robust to using the headline for estimating an article's sentiment.

Table A1

Linear regression models 1c (Stocktwits): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t+x$, with x ranging from 0 to 5. The relevant independent variables include the number of Stocktwits posts (in 100s) and news articles on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,612,172$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	-2.28 ***	-2.21 ***	-2.17 ***	-2.32 ***	-2.37 ***	-2.38 ***
Stocktwits	17.99 **	4.32 *	-1.43	0.46	-0.52	-0.51
News	7.65	2.99	1.57	1.74	3.09.	1.23
Quarterly Report	16.89 ***	3.36 *	-3.08 ***	-3.22 ***	-3.09 ***	-4.19 ***
AV (1st Decile)	-3.89 ***	-0.42	0.79.	0.50	0.44	0.86.
AV (10th Decile)	15.07 ***	3.69 **	-0.43	0.13	-0.03	-0.94
Return (1st Decile)	6.15 ***	1.33.	0.71	-0.53	0.70	-0.50
Return (10th Decile)	4.72 ***	2.68 **	-0.89.	1.18 *	-0.87	-1.42 *
Mean Day	0.85 ***	0.86 ***	0.86 ***	0.86 ***	0.86 ***	0.86 ***
Mean Month	0.94 ***	1.02 ***	1.05 ***	1.04 ***	1.04 ***	1.05 ***
Time Controls				Yes		
R ²	0.158	0.131	0.129	0.129	0.129	0.129

Table A2

Linear regression models 2b (Stocktwits sentiment): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t+x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles, divided by their sentiment, on day t . We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,612,172$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	-2.08 ***	-2.14 ***	-2.14 ***	-2.23 ***	-2.28 ***	-2.28 ***
Stocktwits (bullish)	78.27 ***	10.20	-14.53 **	2.42	-8.25.	-7.60.
Stocktwits (neutral)	-5.12	-3.69	1.95	-5.76 *	-2.70	-3.53
Stocktwits (bearish)	-10.54	22.75	10.92	20.09 *	22.37.	24.70.
News (positive)	-15.18	-9.04	-1.33	-1.62	5.79	5.28
News (neutral)	-2.15	2.40	4.52	3.40	0.39	-1.68
News (negative)	54.88 ***	20.13 *	0.61	3.19	3.23	-0.27
Quarterly Report	19.07 ***	4.56 *	-3.16 **	-2.13 **	-2.96 ***	-3.97 ***
AV (1st Decile)	-5.85 ***	-0.42	1.26 **	0.62	0.87	1.29
AV (10th Decile)	10.38 ***	3.89 **	0.75	0.64	1.07	0.17
Return (1st Decile)	8.08 ***	1.19	0.15	-0.68	0.21	-0.97
Return (10th Decile)	5.28 ***	2.83 ***	-0.92	1.20 *	-0.94	-1.50
Mean Day	0.85 ***	0.86 ***	0.86 ***	0.86 ***	0.86 ***	0.86 ***
Mean Month	0.90 ***	1.01 ***	1.06 ***	1.04 ***	1.05 ***	1.05 ***
Time Controls				Yes		
R ²	0.175	0.132	0.130	0.129	0.130	0.130

Table B1

Linear regression models 1d (NER on title and first five sentences): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t+x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles on day t . This table differs from [Table 4](#) in that additionally the first five sentences of the news articles are searched for company names instead of just the titles. We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to [Newey and West \(1987, 1994\)](#). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	-3.13 ***	-2.91 ***	-2.79 ***	-2.85 ***	-3.00 ***	-3.08 ***
Posts	58.09 ***	12.61	-5.68	-3.08	-0.92	-2.41
News	1.16.	0.38	0.09	0.20	0.31	0.05
Quarterly Report	16.34 ***	3.02.	-3.18 ***	-2.69 ***	-2.78 ***	-4.31 ***
AV (1st Decile)	-1.28.	0.58	1.11 *	0.93 *	0.75	1.02.
AV (10th Decile)	23.85 ***	6.48 ***	-0.60	-0.06	-1.06	-0.76
Return (1st Decile)	3.83 ***	0.35	0.61	-0.70	0.64	-0.50
Return (10th Decile)	4.02 **	3.31 **	-0.56	0.88	-1.61 *	-2.17 **

(continued on next page)

Table B1 (continued)

Day	0	1	2	3	4	5
Mean Day	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.18 ***	1.23 ***	1.26 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.231	0.209	0.208	0.208	0.208	0.208

Table C1

Linear regression models 2c (sentiment of news article titles): The dependent variable is the absolute change in the number of Robinhood users ΔN_{t+x} on day $t + x$, with x ranging from 0 to 5. The relevant independent variables include the number of Reddit posts and news articles, divided by their sentiment, on day t . This table differs from Table 6 in that only the titles of news articles are used for sentiment analysis. We control for all 10 deciles, but report only the top and bottom deciles of the *AV* and *Return* variables for the sake of simplicity. We use robust standard errors according to Newey and West (1987, 1994). $N = 4,692,508$. Significance levels: . $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Day	0	1	2	3	4	5
Intercept	−3.11 ***	−2.91 ***	−2.79 ***	−2.85 ***	−3.00 ***	−3.07 ***
Posts (positive)	75.09 ***	5.57	−6.83	−3.38	−2.78	5.25
Posts (neutral)	48.51 *	17.16	−21.16 *	−16.49 *	−24.62 ***	−21.81 *
Posts (negative)	28.86	26.98	46.35	38.05	76.92 *	27.52
News (positive)	−28.28	−18.52	−7.70	2.59	1.86	1.51
News (neutral)	4.92	3.33	2.75	0.10	0.47	−2.71
News (negative)	41.32 ***	13.83	−2.66	−0.12	0.28	6.97
Quarterly Report	16.29 ***	3.91 *	−2.49 **	−2.63 ***	−2.51 **	−4.73 ***
AV (1st Decile)	−1.34	0.56	1.16 **	0.96 *	0.81	1.05
AV (10th Decile)	23.76 ***	6.49 ***	−0.51	−0.03	−0.9	−0.77
Return (1st Decile)	3.69 **	0.21	0.47	−0.78	0.46	−0.58
Return (10th Decile)	4.29 ***	3.44 ***	−0.55	0.87	−1.62 *	−2.15 **
Mean Day	0.73 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***	0.72 ***
Mean Month	1.18 ***	1.23 ***	1.26 ***	1.25 ***	1.25 ***	1.25 ***
Time Controls				Yes		
R ²	0.232	0.209	0.209	0.208	0.209	0.208

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